Self-Supervised Learning for Video Geometry

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1 Introduction

Recent advances in 3D vision have shifted from per-scene optimization methods like NeRF [1] and Gaussian Splatting [2] to general feed-forward models that can reconstruct geometry across different scenes. VGGT [3] predicts camera poses, depth, and structure from multi-view images in under one second; Geo4D [4] extends to dynamic scenes using video diffusion priors. Both rely on labeled or synthetic supervision, limiting scalability to billions of hours of unlabeled video online.

Videos offer an extraordinarily rich yet underutilized signal for geometric learning. Unlike static images, videos provide: (1) natural multi-view geometry through camera motion, (2) temporal consistency constraints that implicitly encode 3D structure, (3) motion cues that reveal depth ordering and object boundaries, and (4) orders of magnitude higher scale - billions of hours of video exist without geometric labels. Self-supervised learning transformed language [5] and vision [6, 7] by exploiting unlabeled data. Can we do the same for video geometry?

Self-supervised monocular depth has advanced significantly [8, 9], but methods treat video as independent frames. Recent work like Depth Anything 3 [10] achieves state-of-the-art multi-view geometry through teacher-student training with synthetic data and pseudo-labeling. Meanwhile, work like SAM 3D [11] demonstrates impressive single-image 3D reconstruction by training on diverse viewpoints, yet processes each image independently without leveraging temporal continuity. Designing self-supervised objectives that capture how geometry evolves through camera motion - temporal consistency across frames, multi-view geometric coupling - needs further exploration. Which temporal objectives best teach spatiotemporal reasoning? How data-efficient is learning from video sequences versus static multi-view datasets?

2 Background

2.1 Feed-Forward Multi-View Reconstruction

Multi-view reconstruction traditionally requires optimization-based structure-from-motion [12]. Recent work replaces this with feed-forward prediction. DUSt3R [13] pioneered dense point regression from image pairs but requires global alignment for multiple views. VGGT [3] addresses this problem by utilizing camera tokens and alternating attention, processing 1-200+ images jointly. MonST3R [14] performs pairwise tracking for dynamic scenes; Geo4D [4] adapts video diffusion models. All require expensive supervision: DUSt3R/VGGT use structure-from-motion labels, Geo4D uses synthetic data rendered from 3D assets, limiting real-world scalability.

2.2 Self-Supervised Learning (SSL)

Self-supervised learning (SSL) extracts training signals from data structure. Masked autoencoders [6] reconstruct masked patches; DINO [7, 15] matches predictions between augmented views, producing semantic features without labels. VideoMAE [16] extends masking to spatiotemporal tubes. For geometry, self-supervised depth [8, 9] uses photometric consistency between consecutive frames. However, methods predict per-frame depth independently, treating video as image sequences rather than modeling temporal geometric structure. Recent work shows SSL can teach temporal understanding for actions [17], yet whether temporal

SSL objectives can capture evolving geometry and multi-view consistency remains a fundamental question for scaling video geometric understanding.

3 Problem Statement

While SSL has transformed image understanding, its application to video geometry remains limited by several fundamental challenges:

Appearance vs geometry - Geometric learning requires different signals than appearance-based pretraining. While methods like DINO learn semantic features through augmentation invariance, geometry demands precise metric structure and multi-view consistency. Models can minimize photometric loss through surface appearance matching without capturing underlying 3D structure. Recent work [18] shows appearance-based video generation struggles with accurate 2D motion due to pixel synthesis overhead, suggesting geometry-specific objectives are necessary.

Temporal signals and shortcuts - Videos encode geometric constraints through motion parallax and occlusion patterns, yet naive temporal objectives permit trivial solutions - copying pixels across frames or learning optical flow without depth understanding. Which temporal SSL objectives force genuine spatiotemporal geometric reasoning rather than enabling such shortcuts?

Scaling to real video - Current methods use expensive supervision [3] or synthetic data [4], either limiting data scale or real-world generalization. Whether pretraining on real unlabeled video can match these approaches, how much data is required, and which objectives transfer most effectively to downstream tasks remain uncharacterized - yet are essential for scaling video geometric learning.

4 Research Plan

4.1 Research Questions

(1) Which temporal SSL objectives effectively teach geometric structure? Beyond existing approaches like photometric consistency [8] or masked prediction [16], can we design novel objectives that leverage video-specific signals - cycle consistency across longer sequences, multi-view geometric coupling, or predictive temporal reasoning - to force genuine 3D understanding rather than surface appearance matching?

(2) How do different temporal signals interact in multi-objective training? SSL methods combine multiple objectives [7, 16] - temporal consistency, geometric coupling, predictive modeling. Which combinations are complementary versus redundant for geometric learning? Can we identify minimal sets that maximize geometric supervision while maintaining data efficiency?

(3) What are the data efficiency and scaling properties of video pretraining for geometry? How much unlabeled video is required to match supervised methods like [3] or synthetic approaches like [4]? Can we characterize predictable relationships between pretraining data scale and downstream performance, analogous to scaling laws in language models [19], and which objectives transfer most effectively to downstream tasks?

Methodological framework - This work systematically investigates existing SSL objectives across three categories: *temporal consistency* (photometric matching [8], cycle consistency [20]), *geometric coupling* (depth-flow constraints [21], multi-view consistency [3]), and *predictive modeling* (masked spatiotemporal reconstruction [16]). Based on insights from systematic ablations, this research will explore novel extensions where existing objectives prove insufficient for capturing temporal geometric structure. Through controlled experiments isolating objective type, objective combinations, and data scale, this investigation establishes which temporal signals enable genuine geometric reasoning versus appearance shortcuts, and characterizes their scaling properties.

Experiment design - Pretraining uses real unlabeled video from RealEstate10K [22], containing diverse indoor and outdoor scenes with natural camera motion. Evaluation benchmarks include NYU Depth V2 [23] and KITTI [24] for depth estimation, Sintel [25] for optical flow, and standard metrics for camera pose estimation. Comparisons against supervised baselines [3] and synthetic pretraining approaches [4] measure both downstream task performance and data efficiency curves across varying amounts of pretraining data.

Timeline - Year 1 focuses on establishing the research foundation through comprehensive literature review and initial setup of experiments framework. Existing temporal SSL objectives are systematically evaluated on standard benchmarks, with initial exploration of novel video-specific geometric objectives based on our findings. Preliminary experiments assess learned representations on downstream tasks and inform the research direction for subsequent years.

4.2 Expected Contributions

This work establishes systematic understanding of which SSL objectives teach video geometry, how temporal signals interact in multi-objective training, and the data requirements for matching supervised methods. By identifying minimal objective combinations that force genuine geometric reasoning, this research contributes to scalable geometric learning from unlabeled video. Findings would benefit robotics and world modeling systems that require 3D geometric understanding without expensive supervision. Open-source implementations will support reproducibility and future research in data-efficient video geometric learning.

I would like to pursue this research under the supervision of Dr. Elliott (Shangzhe) Wu, whose VGG background and expertise in physics-grounded representations from raw video data provides good guidance for this research.

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